HABIB UNIVERSITY

Data Structures & Algorithms

CS/CE 102/171 Spring 2025

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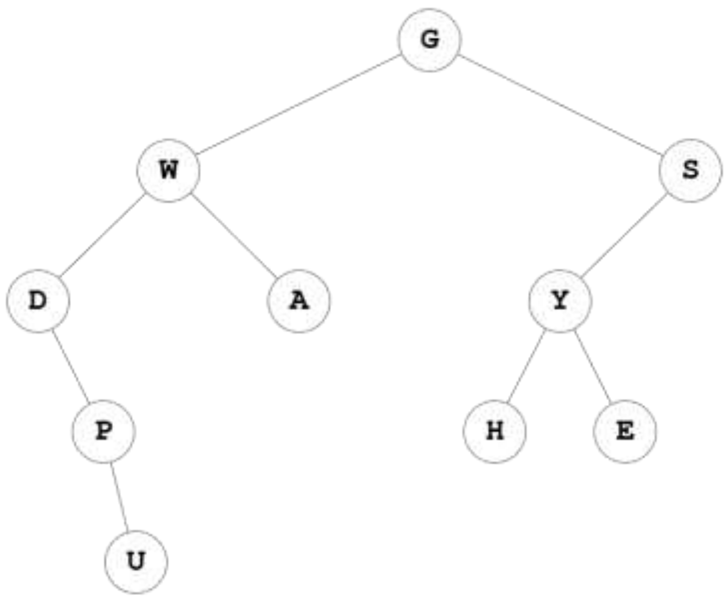
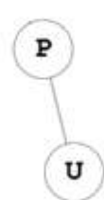
Quiz # 4 Solution

QUESTION: Tree Terminology and Functions

For the given tree, T, answer the following questions:

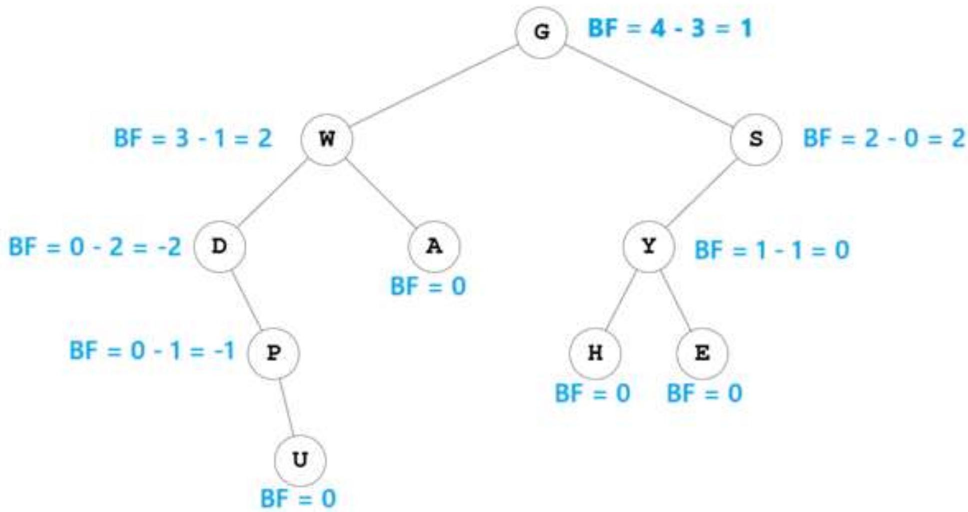


1. degree (P): ***2***
2. depth (T, H): ***3***
3. height (T, W): ***4***
4. Path from W to H: ***Not possible***
5. List down all the leaf nodes: ***[A, U, H, E]***
6. List down all the internal nodes: ***[G, W, S, D, Y, P]***
7. Draw the right subtree of tree rooted at D:



8. Siblings of H: ***E***
9. Cousins of D: ***Y***

10. Calculate Balance Factor of the given tree, by showing each calculation step to get complete grade. You may solve it directly on the tree given in the question.



11. Is this a balanced binary tree? Give reason for your answer: ***No, because multiple nodes have a balance factor is out of range of -1, 0 and 1***

12. Is it Left-Heavy or Right-Heavy or a Perfectly Balanced Binary Tree? Briefly explain your choice. ***Neither, because it was never balanced to begin with***

13. Postorder Tree Traversal: ***{ U, P, D, A, W, H, E, Y, S, G }***

14. Inorder Tree Traversal: ***{ D, P, U, W, A, G, H, Y, E, S }***

15. Array Based Representation of the given tree: **[G, W, S, D, A, Y, None, None, P, None, None, H, E, None, None, None, None, None, U, None, None, None, None, None, None, None, None, None, None, None, None]**

16. Using the Array Representation for Trees, calculate the following: (Show complete working using formulas to prove your answer. Just giving the final answer will result in 0 grade)
    - Parent of Node P

      Position of parent of Node P $= \left\lfloor \frac{f(P)-1}{2} \right\rfloor = \left\lfloor \frac{8-1}{2} \right\rfloor = \left\lfloor \frac{7}{2} \right\rfloor = \lfloor 3.5 \rfloor = 3$

      Parent of Node P = Array[3] = D, which is indeed the parent of Node P

    - Left child of Node U

      Position of left child of Node U = 2 f(U) + 1 = 2(18) + 1 = 37

      This is beyond the range of array index, which means U is a leaf node without any children

***** For the following 3 questions (Q17, Q18 & Q19), do NOT just write down a number as your answer, but also specify the formula that you used to get the answer. No marks will be awarded for giving ONLY the answer *****

17. Maximum number of nodes at level = 4 in a binary tree is: $2^L = 2^4 = 16$

18. A binary tree with 35 nodes will have a minimum height as:

    $$\log_2(n+1) = \log_2(35+1) = \log_2 36 = 5.17 \cong 6$$

19. For a Binary Tree containing 10 internal nodes, the total number of nodes in the entire tree is:

    $2n_i + 1 = 2(10) + 1 = 21$